Lab 1: Requirements Engineering & UML Use-Case Modelling
PROBLEM STATEMENT #37
Group Travel Expense Simplification Engine
SRN:PES1UG24AM462
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AIML SEC-H

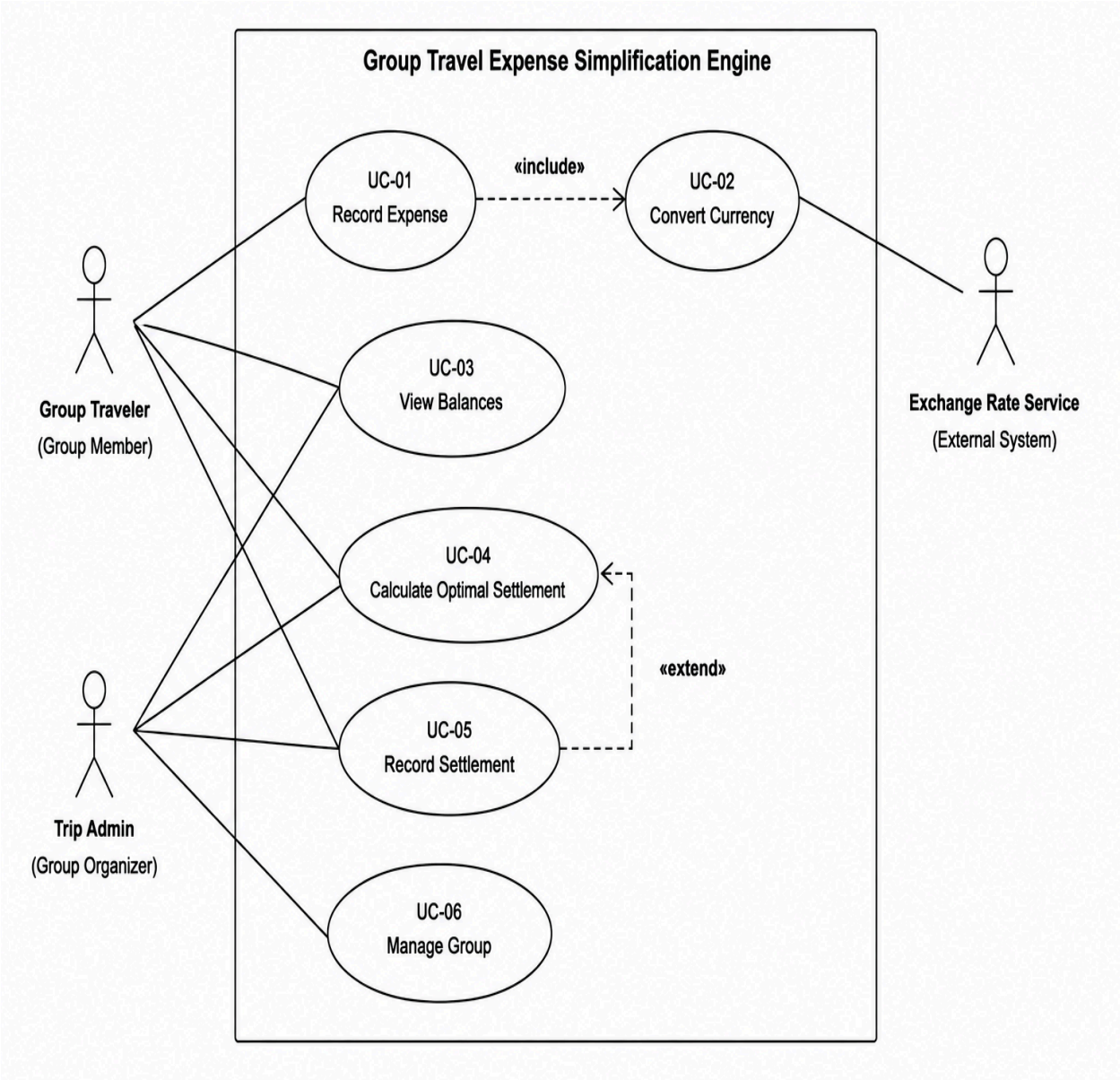
1. **Requirements table:**

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.	High	Pass: Group balance settles to net zero with minimal payment edges. Fail: Sum of debts does not equal sum of credits.	Minimizes the number of transactions required to settle group expenses.
FR-002	Functional	The system shall convert expense amounts from foreign currencies into the selected settlement currency using the applicable daily exchange rate.	High	Pass: Every supported foreign-currency expense is converted using the exchange rate for the expense date and the converted amount is displayed correctly. Fail: Any conversion uses an incorrect or unavailable rate without reporting an error.	Required to accurately combine expenses made in multiple currencies.

FR-003	Functional	The system shall allow a Group Traveler to record an expense with its amount, currency, payer, and participating travelers with their respective shares.	High	Pass: A valid expense is stored with payer, amount, currency, and participant shares, and the shares sum to the expense amount. Fail: The system saves an expense with missing required information or mismatched shares.	Core functionality for accurately tracking unequal group expenses.
FR-004	Functional	The system shall calculate and display each traveler's net balance based on all recorded group expenses and currency conversions.	High	Pass: For every traveler, the displayed balance equals total amount paid minus total amount owed after currency conversion. Fail: Any traveler's displayed balance does not match the calculated value.	Travelers need to know whether they owe money or should receive money.
FR-005	Functional	The system shall allow a Group Traveler to record a completed settlement transaction between two travelers.	Medium	Pass: A valid settlement records the payer, receiver, amount, and status, and the corresponding balances are updated. Fail: A settlement with invalid or missing details is recorded.	Allows the system to track whether suggested debts have actually been settled.
NFR-001	Non-Functional	The settlement calculation engine must execute in under	High	Pass: Benchmarking tests show settlement calculations	Ensures fast settlement calculations even for

		100 ms for groups of up to 50 travelers with 500 expenses.		completing in less than 100 ms for a test case containing 50 travelers and 500 expenses. Fail: Execution time reaches or exceeds 100 ms.	relatively large travel groups.
NFR-002	Non-Functional	The system shall protect stored expense and settlement information so that only authorized members of the corresponding travel group can access it.	High	Pass: An authorized group member can access the group's records, while an unauthorized user is denied access. Fail: An unauthorized user can retrieve or modify group expense or settlement data.	Travel expense records contain private financial information and must be restricted to authorized users.

2. UML Use-Case Diagram



3. **Use-Case Flow Specification**

Use Case: Calculate Optimal Settlement

Main Success Scenario:

1. Group Traveler selects the "Calculate Optimal Settlement" option. System retrieves all recorded group expenses.
2. System converts foreign-currency expenses into the selected settlement currency using the applicable daily exchange rates.
3. System calculates the net balance of each traveler.
4. System separates travelers into debtors and creditors.
5. System applies the min-flow algorithm to calculate the optimal debt settlement.
6. System generates the minimum-transaction settlement graph.
7. System verifies that the total debts equal the total credits.
8. System displays the recommended peer-to-peer payment transactions.
9. Use case ends successfully.

Alternate Flow

2a. Exchange Rate Unavailable

● **2a1.** System displays an error message indicating that the required exchange rate is unavailable.
● **2a2.** User is prompted to retry retrieving the exchange rate.
● **2a3.** If the exchange rate remains unavailable, the settlement calculation is paused and the user is notified.